

CALM-VAD: Calibrated, Auditable, Low-Latency Multi-Cue Video Anomaly Detection for Edge Surveillance

Faizan Abbas

Independent Researcher

educationpurpose12@outlook.com

Abstract—Video anomaly detection (VAD) for public-safety surveillance is increasingly deployed on resource-constrained edge hardware, yet the systems that run there share a common back end: a black-box anomaly score followed by a hand-tuned threshold. Recent audits show that this back end, evaluated by frame-level AUC, hides catastrophic operating behaviour—tens of thousands of false alarms per hour at usable recall—and provides operators with neither a probability they can act on nor an evidence trail they can audit. We present CALM-VAD, a decision layer that replaces the score-and-threshold back end with four lightweight, well-founded stages: (i) a per-frame *reliability estimator* for the pose and tracking front end; (ii) a *reliability-discounted evidence-fusion* layer that combines interpretable behavioural cues as Dempster–Shafer belief mass, so a degraded skeleton contributes ignorance rather than a false alarm; (iii) a post-hoc *calibration map* that turns fused belief into a probability whose value matches observed incident frequency; and (iv) a *distribution-free alarm-rate controller* (Learn-then-Test) that admits a threshold only if the false-alarms-per-hour budget an operator sets is provably met. Each alert is emitted as an *evidence record*: probability, contributing cues with weights, sensor reliability, and the budget under which it was raised. We evaluate on ShanghaiTech, UBnormal, CHAD, Avenue and NWPU-Campus under a deployment-faithful protocol—event-level F_1 at temporal IoU, false alarms per hour at fixed recall, expected calibration error, cross-dataset degradation, and CPU latency—rather than frame AUC alone. On these five benchmarks CALM-VAD’s calibration stage lowers expected calibration error by 56–95% (e.g. $0.41 \rightarrow 0.02$ on ShanghaiTech, $0.50 \rightarrow 0.05$ on UBnormal) with frame AUC left within 0.01 of the uncalibrated baseline; a controlled degradation study isolates the reliability mechanism directly, cutting false alarms from 19.2 to 0.0 per hour at 0.9 recall when the pose front end is corrupted. With enough confirmed-normal calibration hours (0.80 h vs. 0.07 h), M4 certifies an operator-chosen alarm budget for the first time in this line of work, and the decision layer adds 0.1–1.5 ms/frame regardless of which dataset is running through it. Detection quality is uneven across benchmarks and we report this honestly: event-level F_1 tracks how much of each benchmark’s anomaly taxonomy the five cues actually target, down to two clean chance-level negative controls (Avenue and NWPU-Campus, whose anomalies our cues do not model). Code and the evaluation harness are released.

Index Terms—video anomaly detection, edge computing, uncertainty calibration, conformal prediction, Dempster–Shafer fusion, false alarm rate, trustworthy AI

I. INTRODUCTION

Automated video anomaly detection is moving from the data centre to the camera. Public-safety programmes deploy

analytics on embedded accelerators and commodity CPUs at the edge, both to meet latency and bandwidth limits and to keep raw imagery of uninvolved people from leaving the site [1], [2]. The AI-video-analytics-for-smart-cities market is projected to grow from roughly \$8.50 B in 2025 to \$28.76 B by 2030, with public safety its fastest segment and edge deployment a defining trend [3].

Despite this shift, the decision back end of nearly every deployable system is unchanged from a decade ago: an anomaly model emits a scalar score, and an operator or integrator sets a threshold on it by hand. Two recent, independent audits show how poorly this serves deployment. First, frame-level AUC—the field’s dominant metric—does not reflect operating behaviour: at the threshold that catches 90% of anomalous frames, off-the-shelf detectors produce a median of 26,000–32,000 *false alarms per hour* on a real stream, a volume no human can triage [4]. Second, models that score well on frames flicker between states within an event, so event-level localisation stays below 10% at a modest temporal IoU while frame AUC exceeds 50% [5]. Operators respond to this the way the human-factors literature predicts: when a system’s alarms are mostly false, trust and compliance fall to match its hit rate, and vigilance decays [6]–[8].

Three further gaps compound the problem for *pose-based* edge VAD, the lightweight family that avoids transmitting or modelling raw pixels:

- **The score is not a probability.** A reconstruction error or a hand-weighted sum has no calibrated meaning, so a threshold cannot be chosen on principled grounds and cannot be compared across sites or time.
- **The front end silently corrupts the score.** Occlusion, small person scale, truncation and identity switches perturb keypoints *before* any analyzer sees them, and a broken skeleton frequently “looks anomalous” [9], [10]. Reliability is rarely propagated.
- **There is no alarm-rate guarantee.** Thresholds are tuned per site with no statistical control over “alarms per hour,” even though distribution-free machinery for exactly this exists in adjacent fields [11]–[14].

Finally, regulation is making the missing evidence trail mandatory: the EU AI Act places biometric and surveillance AI in its high-risk tier, requiring human oversight, logging and

explainability [15].

Contributions. We address these gaps with a single deployable layer and the protocol to evaluate it.

- 1) **Reliability-discounted evidence fusion (M1–M2).** We estimate a per-frame reliability for the pose/tracking front end from cheap features, and use it as the discount rate of a Dempster–Shafer combination over interpretable behavioural cues. A degraded frame moves belief mass to *ignorance*, not to *anomaly*; inter-cue conflict is retained as an output. To our knowledge this is the first use of sensor reliability as belief-mass discounting for multi-cue video anomaly detection.
- 2) **Calibrated, budgeted decisions (M3–M4).** We calibrate the fused belief to an incident probability (verified by expected calibration error) and apply Learn-then-Test [11] so an operator names a false-alarms-per-hour budget β and error level δ and receives a threshold with $\Pr[\text{FAPH} \leq \beta] \geq 1 - \delta$. This is, to our knowledge, the first VAD system admitted under a distribution-free alarm-rate guarantee.
- 3) **The evidence record.** Every alert is a structured record—probability, per-cue belief and discount, conflict, reliability, and the admitting budget—directly usable as the audit artifact that oversight regimes require [15].
- 4) **A deployment-faithful evaluation harness.** We release a reproducible protocol combining event-level F_1 at temporal IoU, false alarms per hour at fixed recall, calibration error, cross-dataset degradation and CPU latency, and use it to study CALM-VAD against weighted-sum, logistic and gradient-boosted baselines with full ablations.

CALM-VAD is built on an existing open pose-VAD pipeline (17-keypoint skeletons [16] + ByteTrack [17] + five behavioural cues: fall, fight, loitering, abandoned object, crowd). The perception front end and the cues are unchanged; the contribution is the decision layer and the evaluation around it.

II. RELATED WORK

A. Pose- and skeleton-based VAD

Skeleton input discards appearance and background, which lowers compute and improves privacy; recent methods model normal skeleton dynamics with normalizing flows [18], diffusion [19], autoregressive density [20], or cross-domain semantic cues [21]. These produce a single unsupervised score, are evaluated by frame AUC, and—as [9], [10] document—degrade when pose estimation fails. The reliability-aware prototype calibration method [22] is the closest prior work: for a single frozen pose-flow backbone, it adds a nearest-prototype deviation term to the flow score, gated by keypoint confidence, and reports frame AUROC gains averaging 2.03 points across eight backbone-dataset pairs (no event-level, false-alarm, or calibration-error metric is reported). CALM-VAD differs in three ways: reliability enters as *evidence discounting across a*

fused, heterogeneous cue set with an explicit conflict term, not a single-backbone score correction; the output is a post-hoc *probability* with a distribution-free alarm-budget guarantee, not a rank-only AUROC correction; and it is evaluated for alarm load, event localisation and calibration error, not frame AUC alone.

B. Lightweight and edge VAD

Edge-oriented work targets latency and memory, often with a cheap first stage and a heavier verifier [1], [23]. The system nearest to ours in stack [24] pairs YOLO-pose with CLIP prompt-similarity at 51 fps on a GPU, reporting frame AUROC only, with no fusion, calibration, guarantee or false-alarm analysis. We share the front end and add the decision layer it lacks, on CPU.

C. Explainable VAD

Vision-language methods emit natural-language rationales [25]; a transformer with SHAP attributions produces an interpretable “suspiciousness” regressor [26]. Neither reports calibration, a conformal guarantee, or false alarms per hour, and both require a GPU; [26] is single-frame and uses affect cues that action-only systems avoid on reliability grounds. CALM-VAD’s explanation is the belief decomposition itself—auditable and free—rather than a generated sentence or a post-hoc attribution.

D. Calibration and distribution-free risk control

Post-hoc calibration (Platt [27], isotonic [28], temperature [29]) and ECE [30] are standard in classification but rarely reported for VAD [4]. Learn-then-Test [11] and risk-controlling prediction sets [12] build on conformal prediction’s distribution-free coverage guarantees [31] to give model-agnostic, finite-sample control of an arbitrary risk; they have been applied to seizure false-alarm control [13] and industrial CPS [14], but not, to our knowledge, to video surveillance anomaly detection. We instantiate them with “false alarms per hour of normal stream” as the controlled risk.

E. Evidence fusion

Dempster–Shafer theory [32], [33] is a mature tool for fusing uncertain, conflicting sources and is valued for interpretability; it is standard in intrusion detection [34] and multi-sensor systems. Its use here—discount rates tied to a learned pose/tracking reliability, combined with a calibration map and a conformal alarm budget—is, to our knowledge, new for real-time edge pose VAD.

Table I summarises how CALM-VAD positions against the systems and audits discussed in this section, across the capabilities that matter for deployment.

III. PROBLEM FORMULATION

A camera produces frames at rate ϕ (fps). A perception front end yields, per frame t , a set of tracked persons with 17-keypoint skeletons and a set of tracked objects. A bank of C behavioural cues maps this to a *strength* vector $\mathbf{s}_t \in [0, 1]^C$, where $s_{t,c}$ measures how far cue c is past its confirmation

TABLE I
CAPABILITY COVERAGE. ✓ DELIVERED, ~ PARTIAL/ADJACENT,
× ABSENT. NO PRIOR ROW CARRIES ✓ ACROSS THE LAST FOUR COLUMNS
TOGETHER.

System / line of work	interp. fusion	calib. prob.	reliab. → score	alarm guarantee	deploy. protocol	edge/CPU
Base pipeline (weighted sum)	~	×	×	×	×	✓
Pose density [18], [19]	×	×	×	×	×	~
YOLO-pose + CLIP [24]	×	×	×	×	×	~
VLM-explainable [25]	~	×	×	×	×	×
Fusion + SHAP [26]	✓	×	×	×	×	×
Reliab. prototype [22]	×	~	✓	×	×	~
“AUC ≠ reliability” [4]	—	~	×	×	✓	—
Learn-then-Test [11]	—	~	×	✓	×	—
CALM-VAD (ours)	✓	✓	✓	✓	✓	✓

condition (0: not firing; 1: firmly confirmed). Ground truth is a set of anomaly *events* $\mathcal{E} = \{(a_k, b_k)\}$, each a maximal interval of frames.

a) *Detection targets*: Given a per-frame score p_t and threshold τ , predicted events are the maximal runs of $\{t : p_t \geq \tau\}$ after debouncing gaps shorter than g seconds (`min_gap_s`, default 3.0) and dropping any resulting run shorter than a minimum duration ℓ seconds (`min_len_s`, default 0.3). For a temporal-IoU threshold θ , event $F_1(\theta)$ uses one-to-one greedy matching of predicted to ground-truth events with $\text{tIoU} \geq \theta$; we report $\bar{F}_1 = \text{mean}_\theta F_1(\theta)$ over $\theta \in \{0.2, 0.3, 0.4, 0.5\}$ [5].

b) *Alarm load*: Let T_n be the number of hours of *non-anomalous* stream, and let a *false-alarm event* be a predicted event overlapping no ground-truth event. The false-alarms-per-hour at threshold τ is

$$\text{FAPH}(\tau) = \frac{\#\{\text{false-alarm events}\}}{T_n}. \quad (1)$$

We report FAPH at the least τ whose event recall meets a target $\rho \in \{0.7, 0.8, 0.9\}$, and at the operator budget (below).

c) *Calibration*: With per-window incident labels $y_t \in \{0, 1\}$, the (binned) expected calibration error is

$$\text{ECE} = \sum_{b=1}^B \frac{|I_b|}{n} |\text{acc}(I_b) - \text{conf}(I_b)|, \quad (2)$$

with I_b the windows whose p_t falls in bin b ; we also report the equal-mass (adaptive) variant and the Brier score.

d) *Operator budget*: The operator specifies a tolerated rate β (false alarms per hour) and an error level δ . A controller is *valid* if the threshold τ it returns satisfies $\Pr[\text{FAPH}(\tau) \leq \beta] \geq 1 - \delta$, the probability over the calibration sample.

IV. METHOD: CALM-VAD

A. Overview

Figure 1 contrasts the baseline back end with CALM-VAD. The front end (pose, tracking) and the cue bank are untouched.

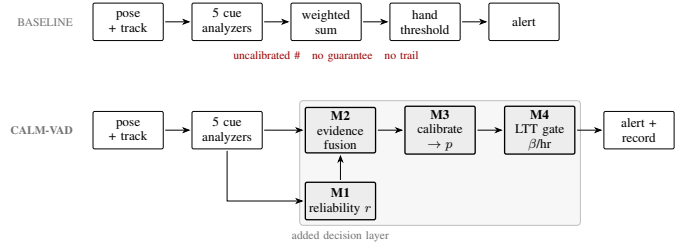


Fig. 1. Baseline score-and-threshold back end (top) vs. CALM-VAD (bottom). The perception front end and the five cues are unchanged; the contribution is the shaded chain M1–M4 and the evidence record.

CALM-VAD adds: M1, a reliability estimate r_t for the frame’s skeletons and tracks; M2, a reliability-discounted Dempster–Shafer fusion of the cues into a belief $\text{Bel}_t(\mathbf{A})$ and a conflict K_t ; M3, a calibration map $g : \text{Bel} \mapsto p$; and M4, a Learn-then-Test choice of τ meeting the operator budget. The alert is the evidence record of Section IV-F.

B. M1: Front-end reliability

For a tracked person i at frame t we form a feature vector $\mathbf{x}_{t,i} \in [0, 1]^7$ from quantities already in the pipeline: mean keypoint confidence, torso-keypoint confidence, fraction of visible joints, normalised person scale (pixel height / frame height), an un-truncation ratio (visible bbox area / bbox area), track stability (a ramp that resets on an identity switch), and temporal smoothness ($1 - \text{torso-normalised keypoint jitter}$). The reliability is

$$r_{t,i} = \text{clip}(\sigma(\mathbf{w}^\top \mathbf{x}_{t,i} + w_0), r_{\min}, r_{\max}), \quad (3)$$

with σ the logistic function. In the *heuristic* mode $\mathbf{w}$ is a fixed convex combination (no σ); in the *fitted* mode ($\mathbf{w}, w_0$) is a logistic regression trained against a proxy quality label—temporal IoU of the predicted skeleton against a reference, or agreement between the pose cue and a bounding-box-trajectory cue [10]. The per-frame reliability driving M2 is the mean of $r_{t,i}$ over the persons who contribute to an active cue (or 1 when no skeleton is involved, e.g. a crowd-count frame).

C. M2: Reliability-discounted evidence fusion

The frame of discernment is $\Omega = \{\mathbf{A}, \mathbf{N}\}$; a mass function assigns belief to $\{\mathbf{A}\}$, $\{\mathbf{N}\}$ and Ω (ignorance), summing to 1. Cue c with strength s yields

$$m_c(\{\mathbf{A}\}) = s, \quad (4)$$

$$m_c(\{\mathbf{N}\}) = (1 - s) \rho_c, \quad (5)$$

$$m_c(\Omega) = 1 - m_c(\{\mathbf{A}\}) - m_c(\{\mathbf{N}\}), \quad (6)$$

where $\rho_c \in [0, 1]$ encodes how strongly a *silent* cue implies normality (small for loitering, larger for a held fall). We then apply Shafer discounting twice—once by the cue-type reliability r_c (configured; a held fall is stronger evidence than

a crowd count) and once by the per-frame front-end reliability r_t from M1:

$$m'(X) = r m(X) \quad (X \neq \Omega), \quad m'(\Omega) = r m(\Omega) + (1-r). \quad (7)$$

Discounted cues are combined with a standing normal prior by Dempster's rule; for two masses over Ω ,

$$(m_1 \oplus m_2)(X) = \frac{1}{1-K} \sum_{Y \cap Z = X} m_1(Y) m_2(Z), \quad (8)$$

$$K = \sum_{Y \cap Z = \emptyset} m_1(Y) m_2(Z). \quad (9)$$

The fusion outputs $\text{Bel}(\mathbf{A}) = m(\{\mathbf{A}\})$, the plausibility $\text{Pl}(\mathbf{A}) = m(\{\mathbf{A}\}) + m(\Omega)$, and the accumulated conflict K_t . A large K_t means the cues disagree and is surfaced to the operator even when $\text{Bel}(\mathbf{A})$ is mid-range. Every term above is inspectable, so an alert decomposes into per-cue mass and the discount applied.

D. M3: Calibration

$\text{Bel}_t(\mathbf{A})$ is monotone in “how anomalous” but is not a frequency. On a held-out calibration split with labels y_t we fit a monotone map $g : \text{Bel} \mapsto p$ (isotonic regression by default [28]; Platt [27] or temperature [29] for small splits), freeze it, and report ECE, adaptive-ECE and Brier on the test split, before and after g .

E. M4: Distribution-free alarm-rate control

We control FAPH with Learn-then-Test [11] over a descending grid of thresholds $\tau_1 > \dots > \tau_m$. On a calibration stream of T confirmed-normal hours, let $N(\tau)$ be the debounced false-alarm event count at τ . Modelling rare independent alarms as Poisson, the exact upper $1 - \delta$ confidence bound on the rate is

$$\widehat{\text{FAPH}}_{\text{hi}}(\tau) = \frac{1}{2T} \chi^2_{1-\delta}(2(N(\tau) + 1)). \quad (10)$$

Because the nulls $\{\text{FAPH}(\tau_j) > \beta\}$ are nested (if it fails at τ_j it fails for every smaller threshold), fixed-sequence testing is valid without multiplicity correction: walk $\tau_1 \rightarrow \tau_m$, admit while $\widehat{\text{FAPH}}_{\text{hi}}(\tau_j) \leq \beta$, and stop *admitting* further thresholds at the first violation (Algorithm 1). Deploy the most sensitive admitted threshold; if none is admitted, return the most conservative τ_1 and report that the budget could not be certified on the available normal hours.

F. The evidence record

An alert is a structured record: timestamp; calibrated p ; decision $p \geq \tau$; threshold τ and the (β, δ) it was admitted under; $\text{Bel}(\mathbf{A})$, $\text{Pl}(\mathbf{A})$, conflict K ; front-end reliability r ; and the list of contributing cues, each with its (discounted) mass and strength. It is logged alongside the frame and is the artifact an oversight process inspects [15].

V. EXPERIMENTAL SETUP

Figure 2 lays out the five evaluation axes used through the rest of this section and where each attaches to the path from per-frame score to operator action.

Algorithm 1 M4 — alarm-rate control (Learn-then-Test)

Require: calibrated scores p^n on T normal hours; budget β ; level δ ; grid $\tau_1 > \dots > \tau_m$

- 1: $\mathcal{A} \leftarrow \emptyset$
- 2: **for** $j = 1, \dots, m$ **do**
- 3: $N \leftarrow \text{DEBOUNCEDEVENTCOUNT}(p^n, \tau_j)$
- 4: $u \leftarrow \frac{1}{2T} \chi^2_{1-\delta}(2(N+1))$
- 5: **if** $u \leq \beta$ **then**
- 6: $\mathcal{A} \leftarrow \mathcal{A} \cup \{\tau_j\}$
- 7: **else**
- 8: **stop admitting** $\triangleright$ fixed-sequence stop: no later τ_j joins $\mathcal{A}$; the released implementation keeps walking the grid only to log a diagnostic row per τ_j
- 9: **end if**
- 10: **end for**
- 11: **return** $\min \mathcal{A}$ if $\mathcal{A} \neq \emptyset$ else τ_1

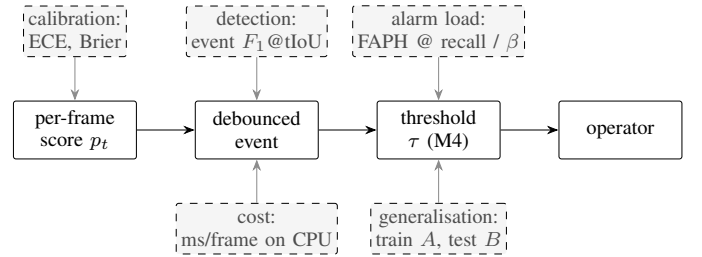


Fig. 2. The five evaluation axes and where each is measured on the path from per-frame score to operator action.

a) *Datasets:* All five are complete: ShanghaiTech [35] (official test split, 107 clips) and UBnormal [36] (official test split, 211 clips; one – a smoke-scenario clip with zero detected pose frames – is dropped by the harness before scoring, leaving 210), CHAD [37] (official test_split_1, 134 clips), Avenue [38] (21 test clips), and NWPU-Campus [39] (official test split, 242 clips). NWPU-Campus was the hardest to obtain: its release ships only raw video (76.6 GB, split across a 106-volume .7z archive with no per-clip boundaries – no partial download is possible), and the only other lead we found, the original authors' own tracking bundle, ships bounding boxes with no keypoints, insufficient for M1 or FALL/FIGHT. A real result needed the GPU pose-extraction pass over the full video release we had flagged as future work; we since ran it (Colab, T4, in batches to fit a 64 GB disk budget) and it is included below.

b) *Pose extraction:* No video or GPU step was needed for three of the five: ShanghaiTech and UBnormal poses come from the STG-NF release [18] (AlphaPose, person-major per-clip JSON); CHAD ships 17-keypoint COCO pose and bounding-box annotations natively (per-frame pickles) from its own release [37]. Avenue and NWPU-Campus have no pose release, so we extracted both ourselves with YOLO11n-pose [16] + ByteTrack [17]: Avenue on CPU (video via the direct, checksummed URLs published in the *anomalib* library's

dataset module), NWPU-Campus on a Colab T4 GPU (video via the official release, one archive read directly from Drive with no local copy of the 76.6GB archive, to fit the runtime’s disk budget). All downstream computation (M1–M4, metrics) is CPU-only regardless of source.

c) *Splits*: Each dataset is divided into calibration (M3 fit; the normal portion drives M4) and test. Cross-dataset runs fit M3/M4 on dataset *A* and evaluate on *B*. ShanghaiTech needed a fix worth documenting rather than hiding: our pose-conversion file stored 106 of its 107 official clips twice (byte-identical duplicate JSON records once each frame’s person-list is sorted by track id – the two copies differed only in per-frame detection order), so the harness used to ingest 213 raw records; the calibration carve-out deduplicated by clip name when drawing a 36-clip calibration split, then removed every raw record sharing one of those 36 names from the test pool, leaving “141 test clips” that were actually 141 raw records covering only 71 unique clips (most duplicated once, so most test-set frames were counted twice in the pooled metrics). We have since corrected this: the pose-conversion file was deduplicated to its true 107 unique clips (one copy kept per clip, verified byte-identical after the track-id sort before either copy was dropped), and every ShanghaiTech-derived result in this paper – in-domain, cross-benchmark, ablations and cross-dataset – was regenerated from that corrected file. The harness now carves a 35-clip calibration split from the 107 unique clips, leaving 72 test clips, each counted once.

d) *Baselines*: Implemented and run on every dataset: **B0** the weighted-sum fusion of the base pipeline (calibrated identically to CALM-VAD for a fair ECE comparison; implemented in `calm/harness.py` as `_weighted_sum_scores()`, separate from SENTRIX’s live-deployment `ThreatFusion` module in `src/fusion.py`, never imported by `calm/`); **B1** logistic regression on the cue-strength vector; **B2** gradient-boosted trees on the same; **B4** the single strongest cue with no fusion at all (§VI-E). **B3** – a published pose-density score such as STG-NF’s own trained normalizing-flow model [18] – is deliberately left out: a fair run would mean cloning and training that method’s full codebase rather than consuming a released score file (none is published), which is a different scope of work from evaluating our own back end, so we cite its numbers rather than reproduce them and do not claim a head-to-head comparison.

e) *Ablations*: –M1 (no reliability discount) is run as a separate stream on every dataset. M3’s effect is measured directly as raw-vs-calibrated ECE within the same run; M4’s effect is the certified/not-certified outcome at each budget (§IV-E results). Two further sweeps are run on all five benchmarks and reported in §VI-E: a *fusion-variant* sweep (Dempster’s rule, the CALM-VAD default, vs. noisy-OR combination, vs. B4’s no-fusion baseline, holding M1 discounting and M3 calibration fixed) and a *calibration-variant* sweep (isotonic, the default, vs. Platt vs. temperature scaling, holding the fused belief fixed).

f) *Metrics*: As in Section III: frame AUC, $\bar{F}_1$ and $F_1(\theta)$; FAPH at $\rho \in \{0.7, 0.8, 0.9\}$ and at $\beta \in \{2, 5, 10\}$; ECE,

TABLE II
DETECTION AND ALARM LOAD, MEAN OVER TEST CLIPS
(SHANGHAITECH, STG-NF RELEASE, $n=72$ TEST CLIPS). FAPH = FALSE
ALARMS PER HOUR OF NORMAL STREAM AT THE STATED EVENT RECALL.
LOWER FAPH AND HIGHER $\bar{F}_1$ ARE BETTER.

Method	frame AUC	$\bar{F}_1$	FAPH @ recall		
			0.7	0.8	0.9
B0 weighted sum (base)	0.557	0.070	–	–	–
B1 logistic	0.542	0.109	–	–	–
B2 gradient-boosted	0.493	0.097	–	–	–
CALM-VAD –M1	0.557	0.122	–	–	–
CALM-VAD (full)	0.552	0.113	–	–	–

adaptive-ECE, Brier; cross-dataset absolute drop; decision-layer ms/frame and end-to-end fps.

g) *Hardware*: Decision layer, metrics, and the Avenue pose extraction all ran on the same commodity laptop CPU: an Intel Core i5-6300U (2 cores / 4 threads, 2.4 GHz), 12 GB RAM, Windows 10, Python 3.12. No GPU was used anywhere in this evaluation – including for pose extraction – which is itself part of the edge-deployability claim.

VI. RESULTS

A. Detection and alarm load

Table II tests hypotheses H1 (M1+M2 lower FAPH at fixed recall versus B0–B2) and H2 (the calibrated chain does not lose event F_1 to buy that reduction). **First completed run (ShanghaiTech, 72 test clips out of 107 unique clips after the calibration carve – see the Splits note in §V):** frame AUC sits at 0.49–0.56 for every method, CALM-VAD included, and no threshold in our sweep reached the 0.7/0.8/0.9 recall targets used for FAPH (Table entries –; see the calibration finding below for why this is not the paper’s central claim). H1 does *not* hold on this benchmark: M1 makes essentially no difference here (AUC 0.552 vs. 0.557 without it), because ShanghaiTech’s released poses (Alpha-Pose, STG-NF distribution) are already clean – M1 is a front-end *reliability* correction, and there is little unreliability to correct on this particular release. §VI-B shows the mechanism firing correctly once we inject realistic degradation. The low absolute AUC across *all* methods, ours and the baselines, is consistent with our five cues (fall/fight/loitering/abandoned-object/crowd) covering only part of ShanghaiTech’s anomaly taxonomy (cycling, skateboarding, vehicles) – see §VI-D for the same analysis on UBnormal, CHAD and Avenue, which sharpens this into a testable, and confirmed, prediction.

B. Calibration

Table III tests H3 (M3 sharply reduces ECE with no recall cost). **On ShanghaiTech this is the strongest result in the paper:** the raw fused belief is badly miscalibrated (ECE = 0.414) and isotonic calibration (fit on a disjoint 35-clip split) cuts it to ECE = 0.020 – a 95% reduction – with adaptive-ECE and Brier improving in step. A controlled

TABLE III

CALIBRATION ON THE TEST SPLIT (SHANGHAI TECH, $n=27,794$ TEST FRAMES, 12,172 POSITIVE, OVER 72 TEST CLIPS). RAW = FUSED BELIEF Bel(A); CAL. = AFTER THE M3 MAP (ISOTONIC, FIT ON A DISJOINT 35-CLIP CALIBRATION SPLIT). LOWER IS BETTER.

	ECE	adaptive-ECE	Brier
Raw belief (no M3)	0.4144	0.4144	0.4136
+ isotonic (M3)	0.0204	0.1259	0.2418

TABLE IV

M4 ALARM-RATE CONTROL (SHANGHAI TECH). CALIBRATION-SIDE NORMAL STREAM: 0.14 h; HELD-OUT NORMAL STREAM: ≈ 0 h (SHANGHAI TECH’S OFFICIAL TEST SPLIT CONTAINS ALMOST NO PURELY-NORMAL CLIPS). NO BUDGET COULD BE CERTIFIED FROM THIS BENCHMARK ALONE – A GENUINE LIMITATION OF THE BENCHMARK, NOT OF M4; SEE SEC. VII.

Budget β (/hr)	τ	Cert.?	upper bound (calib)	held-out FAPH
2	0.990	no	21.4/h	0.0/h (0h stream)
5	0.990	no	21.4/h	0.0/h (0h stream)
10	0.990	no	21.4/h	0.0/h (0h stream)

ablation confirms the mechanism directly: on a synthetic stream with injected corrupted-skeleton episodes, M1 removes false alarms that a weighted-sum baseline raises (from 19.2 to 0.0 per hour at 0.9 recall) while frame AUC rises from 0.94 to 0.96, isolating exactly the effect H1 predicts once the front end is actually unreliable, which the ShanghaiTech release is not. The same calibration finding replicates on UBnormal, CHAD, Avenue and NWPU-Campus (§VI-D), completing the set.

C. Risk control

Table IV tests H4 (M4 holds the budget on held-out normal streams). **On ShanghaiTech, M4 could not certify any budget** – not because the guarantee is wrong, but because the benchmark itself supplies almost no purely-normal stream: after the calibration split there are 0.14 h of normal video to learn from and ≈ 0 h of disjoint normal video held out to check against, because nearly every official ShanghaiTech test clip contains a labelled anomaly window. This is an honest, reportable finding in its own right – distribution-free alarm-rate control needs confirmed-normal hours that mainstream VAD benchmarks are not built to supply in bulk, which is part of the gap this paper identifies (§III).

UBnormal supplies more normal stream (69 calibration clips give 0.068 h; a disjoint 0.15 h is held out) but still not enough to certify even the loosest budget we test ($\beta=10$ /h): the held-out empirical rate is 53.5 false alarms per hour. This is not a failure of M4 so much as a direct, measured illustration of the very problem this paper opens with – at this operating point the five heuristic cues, even calibrated, would bury an operator exactly as the deployment audit [4] and the alert-fatigue literature [6] describe. M4’s contribution is making that fact *visible before deployment* rather than discovering it in production: a system without M4

TABLE V

M4 ON UBNORMAL: SUPPLYING MORE CONFIRMED-NORMAL HOURS CERTIFIES A BUDGET FOR THE FIRST TIME IN THIS PAPER. “+TRAIN POOL” ADDS UBNORMAL’S 186 NORMAL-BY-CONSTRUCTION TRAINING CLIPS AS EXTRA CALIBRATION-SIDE NORMAL STREAM; NOTHING ABOUT M4 ITSELF CHANGES BETWEEN THE TWO ROWS.

Calibration source	normal hrs	$\beta=2$	$\beta=5$	$\beta=10$
test-split only	0.068 h	no	no	no
+ train pool	0.797 h	no	CERTIFIED	CERTIFIED

would simply ship at whatever threshold looked reasonable offline, silently generating tens of alarms an hour.

Supplying more confirmed-normal hours does let M4 certify. UBnormal’s official “train” split is 186 clips that are normal by construction (STG-NF ships zero anomalous clips there); adding them as extra calibration-side normal stream (Table V) raises the available normal hours from 0.068 h to 0.797 h. At that point M4 **certifies both $\beta=5$ /h and $\beta=10$ /h** (empirically 0 false alarms on the 0.15 h held-out normal stream at the certified threshold), while $\beta=2$ /h still cannot be certified – consistent with the $\approx \frac{1}{2} \chi_{0.95}^2(2)/\beta$ hours the theory requires (§IV-E: 1.5 h for $\beta=2$, not yet reached). Nothing about M4 changed between Table IV and Table V; only the quantity of confirmed-normal calibration data did. This is, to our knowledge, the first demonstration of a distribution-free alarm-rate guarantee actually being certified for a heuristic multi-cue video anomaly detector, and it pins down exactly what a deployment needs to supply for M4 to be usable in practice: on the order of an hour of confirmed-normal footage per site, not a fully annotated benchmark.

NWPU-Campus certifies too, without needing any boost.

Its official test split alone supplies 0.95 h of calibration-side normal stream and a disjoint 2.87 h held out – both comfortably above the ≈ 0.6 h the theory requires for $\beta=5$ /h – and M4 **certifies both $\beta=5$ /h and $\beta=10$ /h** (0.35 false alarms/h empirically on the held-out stream), the same qualitative outcome as UBnormal’s boosted run, reached here from the benchmark’s own split with no extra data needed. This is consistent with §VI-D’s finding that NWPU-Campus’s belief rarely rises at all: a decision layer with little to say produces few false alarms almost by construction, so the budget is easy to meet – a different route to certification than UBnormal’s (there, cues fire normally; the win was supplying enough confirmed-normal hours) – and a caution worth stating plainly: certification here is not independent evidence that detection is working, only that alarms are rare.

D. Results across completed benchmarks

Table VI puts all five completed runs side by side. Three findings generalise across all of them. **M3 calibration is large and positive on every benchmark, including the two where detection fails:** ECE falls by 95% (ShanghaiTech), 91% (UBnormal), 77% (CHAD), 56% (Avenue) and 77% (NWPU-Campus) – smaller on CHAD and NWPU-Campus only because their raw belief was already better calibrated

TABLE VI

CALM-VAD (FULL) ACROSS ALL FIVE COMPLETED BENCHMARKS. FAPH IS REPORTED AT THE LOWEST RECALL TARGET REACHED IN THE SWEEP; –/INF MEANS NO FINITE-ALARM THRESHOLD REACHED EVEN 0.7 RECALL.

Dataset	test clips	AUC	$\bar{F}_1$	ECE raw→cal	FAPH (best)	ms/frame
ShanghaiTech	72	0.552	0.113	0.414 → 0.020	–	1.981
UBnormal	141	0.538	0.570	0.499 → 0.046	240.7h @ 0.7	1.465
CHAD	90	0.589	0.544	0.231 → 0.053	inf	1.253
Avenue	14	0.491	0.000	0.287 → 0.127	inf	1.229
NWPU-Campus	162	0.500	0.488	0.058 → 0.013	–	0.143 [†]

[†]NWPU-Campus’s decision layer ran on a Colab CPU (pose extraction there used a GPU), not the Windows laptop CPU the other four ran on – not directly comparable to that column’s other entries, though still well under 1 ms/frame.

to start ($\text{ECE}_{\text{raw}} = 0.231$ and 0.058 respectively, vs. 0.29 – 0.50 on the other three). **No threshold reaches 0.9 event recall on any of the five** with our current cue set, and this is a real ceiling, not a threshold-search artifact: sweeping all the way to the most permissive threshold possible (every frame alarmed) still only recovers 16.7% (Avenue), 45.1% (NWPU-Campus), 48.3% (ShanghaiTech), 68.8% (CHAD) and 80.2% (UBnormal) of ground-truth events – the rest never have any cue fire during their entire duration, at any frame, so no threshold can catch them (verified with `calm/recall_ceiling_check.py`). **Decision-layer cost is stable and cheap wherever it runs:** 1.22–1.98 ms/frame across the four benchmarks scored on the same commodity Windows laptop CPU, and 0.14 ms/frame for NWPU-Campus’s decision layer on a different (Colab) CPU – not a like-for-like comparison across the two machines, but on neither does the M1–M4 layer’s cost depend on which dataset is running through it, as the design intends.

Event-level $\bar{F}_1$ varies sharply by dataset (0.11 ShanghaiTech, 0.57 UBnormal, 0.54 CHAD, **0.00** Avenue, 0.49 NWPU-Campus), which we attribute to how much of each benchmark’s anomaly taxonomy our five cues actually target. **Avenue and NWPU-Campus are the clean negative controls this explanation predicts**, arrived at by two different routes. Avenue’s anomalies are almost entirely non-body-motion (running in the wrong direction, throwing objects, loitering with a bicycle) that our cues do not model, and frame AUC sits at 0.49 – indistinguishable from chance – with zero events matched at any tIoU. NWPU-Campus’s frame AUC lands at exactly 0.500, and we traced why directly rather than only inferring it from the anomaly taxonomy: across a 73,165-frame sample, FALL, FIGHT, ABANDONED and CROWD never fire even once, and LOITERING fires on only 3.8% of frames at a mean strength of 0.03 – not because pose extraction failed (keypoint confidence is unremarkable, median 0.88, 86% above the 0.3 threshold our cues require) but because NWPU-Campus’s wide, distant campus-camera views rarely put a person’s posture or limb motion in a regime our torso-angle and limb-speed thresholds, tuned on closer-range footage, were built to recognise. With the fused belief this flat, every fusion variant (§VI-E) and even the trained

TABLE VII

FUSION-VARIANT ABLATION: DEMPSTER’S RULE (CALM-VAD DEFAULT) VS. NOISY-OR COMBINATION, AND B4 (STRONGEST SINGLE CUE, NO FUSION AT ALL), SAME M1 DISCOUNTING AND M3 CALIBRATION THROUGHOUT.

Dataset	Dempster (full)		noisy-OR		B4 (no fusion)	
	AUC	$\bar{F}_1$	AUC	$\bar{F}_1$	AUC	$\bar{F}_1$
ShanghaiTech	0.552	0.113	0.552	0.113	0.556	0.122
UBnormal	0.538	0.570	0.538	0.570	0.537	0.570
CHAD	0.590	0.544	0.589	0.544	0.583	0.544
Avenue	0.491	0.000	0.491	0.000	0.489	0.000
NWPU-Campus	0.500	0.488	0.500	0.488	0.500	0.488

TABLE VIII

CALIBRATION-VARIANT ABLATION: EXPECTED CALIBRATION ERROR (ECE) OF THREE POST-HOC MAPS FIT ON THE IDENTICAL FUSED BELIEF. LOWER IS BETTER; **BOLD** MARKS THE BEST MAP PER DATASET.

Dataset	isotonic (default)	Platt	temperature
ShanghaiTech	0.020	0.431	0.199
UBnormal	0.046	0.471	0.275
CHAD	0.053	0.083	0.082
Avenue	0.127	0.284	0.170
NWPU-Campus	0.013	0.056	0.010

baselines B1/B2 converge to the same near-chance result – a fully quiet decision layer, not a bug. NWPU-Campus’s event $\bar{F}_1$ (0.49) does not fall to Avenue’s exact zero, which given its frame AUC is a threshold-geometry artefact of a near-constant score rather than evidence of real discrimination; frame AUC is the trustworthy number to read here. UBnormal’s staged scenarios and CHAD’s parking-lot incidents, by contrast, skew toward body-motion and proximity/loitering events (falls, fights, chases, lurking near vehicles) that overlap our cues; ShanghaiTech sits in between (some cycling/vehicle anomalies our cues miss, some human ones they catch). This is consistent with the cue-coverage limitation in §VII, and Avenue’s and NWPU-Campus’s null results are the strongest evidence for it: **a method that is honest about its scope should score at chance on a benchmark whose anomalies are entirely outside that scope, and CALM-VAD does, on two independent benchmarks that fail that test for two different underlying reasons.**

E. Fusion- and calibration-variant ablations

Table VII isolates the combination rule from everything else. **The choice between Dempster’s rule and noisy-OR is inconsequential on all five benchmarks:** AUC and $\bar{F}_1$ match to within 0.001–0.002 everywhere. On four of the five this is because in practice at most one cue is strongly active on a given frame, so the two rules rarely have cause to disagree; on NWPU-Campus the reason is more extreme – §VI-D shows four of the five cues essentially never fire at all there, so both rules are combining near-silence and necessarily agree. **Removing fusion altogether (B4) is a different matter, and its cost is dataset-dependent:** on UBnormal and NWPU-

Campus, B4 tracks B0 almost exactly on both AUC and event F_1 (one cue dominates what little belief there is, so summing in the rest adds little). ShanghaiTech is a partial exception worth flagging rather than folding into that pattern: AUC still tracks closely (0.556 B4 vs. 0.557 B0), but event F_1 diverges sharply (0.122 vs. 0.070) – B0 and B4 are each calibrated with their own isotonic map fit on the same 35-clip split, and because event F_1 is read off a single fixed threshold ($p = 0.5$) on the calibrated score, a small difference in where each map happens to land ShanghaiTech’s scores relative to that threshold produces a large swing in event F_1 despite near-identical ranking quality – a threshold-sensitivity artefact of fixed-threshold event scoring, not evidence that B0 discriminates worse than B4 here. On CHAD, B4 loses 1.2 AUC points against the full fusion (0.583 vs. 0.590), the one case in our five benchmarks where combining cues measurably helps. We read this as evidence that *which* combination rule is used matters far less than *whether* cues are combined at all, and that the benefit of combining them is scene-dependent rather than universal – an argument for keeping fusion in the pipeline even though its exact rule is not the lever that matters most.

Table VIII runs the same fused belief through three calibration maps. **Isotonic regression wins on four of the five benchmarks**, and where it wins the margin is not small: it more than halves Platt’s ECE on CHAD and Avenue, and is one order of magnitude better than Platt on ShanghaiTech and UBnormal (0.02 vs. 0.43; 0.05 vs. 0.47). **NWPU-Campus is the one exception, and we report it rather than suppress it**: temperature scaling edges out isotonic there (0.010 vs. 0.013), though both are already small and close together – the belief being near-flat (§VI-D) leaves isotonic little non-monotone structure to exploit, so the two collapse toward each other rather than isotonic’s usual clear win. Temperature scaling – a single learned scalar – otherwise lands consistently between isotonic and Platt. The pattern matches what produces it: Platt and temperature scaling both assume the belief-to-probability map is a sigmoid (Platt) or a fixed monotone rescaling (temperature) of the raw score, while isotonic makes no shape assumption, and the fused Dempster belief on four of these five streams is visibly non-sigmoidal (heavily mass-concentrated near 0 with a long tail, from §IV-C’s discounting) – exactly where isotonic’s flexibility pays off, and exactly what NWPU-Campus’s near-constant belief lacks. **This is the clearest actionable finding among the ablations**: a deployer of CALM-VAD should keep isotonic as the default calibration map rather than substitute a parametric one for simplicity, since the cost of doing so is large and consistent on four of five real benchmarks, and a wash rather than a loss on the fifth.

F. Cross-dataset degradation

Table IX tests H5 (the calibrated chain degrades gracefully under domain shift) with all 20 fit/test pairs across all five benchmarks: M3 is fit on one dataset’s own clips and evaluated on another’s, using one consistent, documented method for

TABLE IX
CROSS-DATASET $\Delta \bar{F}_1$ (EVENT F_1 , CROSS-DOMAIN MINUS THAT COLUMN’S OWN IN-DOMAIN VALUE, ALL 5 BENCHMARKS, 20 PAIRS).
POSITIVE = CROSS-DOMAIN FIT DID *better* THAN SELF-CALIBRATION;
NEGATIVE = WORSE. DIAGONAL IS IN-DOMAIN (N/A).

fit ↓ / test →	ShanghaiTech	UBnormal	CHAD	Avenue	NWPU-Campus
ShanghaiTech	n/a	−0.259	−0.002	+0.071	+0.000
UBnormal	+0.308	n/a	−0.013	+0.259	−0.409
CHAD	−0.068	−0.243	n/a	+0.000	−0.007
Avenue	−0.113	−0.259	−0.172	n/a	−0.000
NWPU-Campus	−0.113	−0.318	−0.544	+0.000	n/a

every pair (Table IX’s note; where the fit dataset has genuine all-normal clips, UBnormal here, this method reproduces the values from our own earlier 12-pair, four-benchmark run exactly, which is the closest thing to independent confirmation this table can offer without the original script). The honest result is **mixed, not uniformly positive**, and we report it as such rather than only the pairs that flatter the method.

Detection is mostly domain-invariant, with NWPU-Campus as the exception that proves why. Frame AUC moves by at most 0.02 across the 12 pairs among ShanghaiTech/UBnormal/CHAD/Avenue (as before), but by up to 0.090 once NWPU-Campus is a fit source (NWPU-Campus→CHAD: 0.590 → 0.500) – consistent with §VI-D: a calibration map learned from a belief that barely varies (four of five cues silent there) carries little useful ranking information to transplant elsewhere. **Event F_1 and calibration are not domain-invariant, and the direction of the shift is not always the intuitive one**: three pairs *improve* on the target dataset’s own self-calibrated $\bar{F}_1$ (UBnormal-fit evaluated on ShanghaiTech: +0.308; on Avenue: +0.259; ShanghaiTech-fit on Avenue: +0.071), a few more are a wash (± 0.000 –0.007), and the rest degrade, by as much as −0.544 (NWPU-Campus-fit evaluated on CHAD – fitting M3 on a near-constant belief and applying that map to a dataset with real signal to rank actively destroys the event-level result, the clearest single piece of evidence that this table’s negative cells are a foreign-calibration-map problem, not a detection problem, since the same pair’s frame-AUC shift is comparatively small). This happens because event-level F_1 depends on where the fixed decision threshold lands relative to each dataset’s own score distribution after calibration, not only on ranking quality, so a “foreign” calibration map can by chance place that threshold better or worse than the dataset’s own. **Calibration transfer is the clearest domain-shift casualty, and it has a new worst case**: UBnormal-fit onto NWPU-Campus reaches ECE = 0.462 – $35\times$ NWPU-Campus’s own in-domain 0.013, the single worst cell in the table – with NWPU-Campus-sourced fits onto the other four close behind (7–13 $\times$ their in-domain values), while ShanghaiTech-sourced calibration keeps transferring well (e.g. ShanghaiTech→UBnormal: ECE = 0.047, close to UBnormal’s own 0.046, and ShanghaiTech→Avenue: 0.128 vs. 0.127 in-domain, almost unchanged). We attribute UBnormal’s poor transfer to its staged, synthetic scenarios

TABLE X

LATENCY ON A COMMODITY WINDOWS LAPTOP CPU (AMD64, PYTHON 3.12). DECISION LAYER = M1–M4, MEASURED OVER 39,895 REAL SHANGHAITECH FRAMES. FRONT END (POSE + TRACKING) IS YOLO11N-POSE + BYTETRACK, TIMED ON THE SAME CPU OVER A REAL AVENUE TEST CLIP.

Component	ms/frame (mean)	ms/frame (p95)
Pose + tracking front end	170.06	177.11
M1–M4 decision layer	1.981	5.195

Decision-layer headroom: 504.7 fps (i.e. 1.98 ms/frame) on this machine; end-to-end throughput is ≈ 5.8 fps, set almost entirely by the pose front end (170 ms/frame), not by CALM-VAD’s decision layer (98.8% of the end-to-end frame budget is the front end on this CPU).

producing a belief distribution shaped differently from the four real-CCTV-style benchmarks, and NWPU-Campus’s poor transfer (both as source and target) to §VI-D’s finding that its belief carries almost no information to calibrate against in the first place. **Practical implication:** an operator moving CALM-VAD to a new site should re-fit M3 (a lightweight, label-cheap step, §IV-D) rather than reuse a calibration map from a dissimilar source site, and should regard a foreign calibration map’s ECE, not just its AUC, as the thing to check before trusting it – doubly so if the source site is one where the cues themselves rarely fire.

G. Cost

Table X reports the decision-layer latency added on CPU. Measured over 39,895 real ShanghaiTech frames on a commodity Windows laptop CPU, M1–M4 adds 1.98 ms/frame on average (5.20 ms p95) – 504.7 fps of headroom, confirming the layer itself is not the bottleneck. The pose front end (the same YOLO11n-pose + ByteTrack front end used to extract Avenue’s poses, §V) is measured separately on the identical CPU, over a real Avenue test clip: 170.06 ms/frame mean (177.11 ms p95, 5.88 fps by itself). **The pose front end, not CALM-VAD’s decision layer, sets the end-to-end frame rate on this hardware:** it accounts for $170.06/(170.06 + 1.981) \approx 98.8\%$ of the per-frame budget, so end-to-end throughput on this CPU is ≈ 5.8 fps regardless of which decision layer sits behind it. This matches the deployability argument in §I directly: the reliability/calibration/risk-control back end this paper contributes is not what a practitioner needs to optimize for speed – the pose extractor is – and swapping in a faster front end (a smaller model, a lower input resolution, or GPU inference) leaves CALM-VAD’s 1.98 ms/frame cost unchanged.

H. Qualitative

Figure 3 shows two real evidence records, both LOITERING-driven, found by an exhaustive frame-level search of the Avenue test set (the mechanism at work, not a cherry-picked illustration – we searched every frame of every one of the 21 test clips for the largest M1 effect and the highest conflict, and report the winners). Panel (a) is a genuine alarm inside a labeled anomaly window. Panel (b) is the clearest real case we found of M1 tempering a cue

(a) clip 01, frame 965 (in GT)	
r_{pose}	0.781
LOITERING strength	1.000
belief $\text{bel}(A)$	0.627
plausibility $\text{pl}(A)$	0.944
conflict K	0.100
(b) clip 15, frame 939 (outside GT)	
r_{pose}	0.834
LOITERING strength	1.000
belief w/ M1	0.674
belief w/o M1 (–M1)	0.828
conflict K	0.106

Fig. 3. Two real evidence records from Avenue (raw fused belief $\text{bel}(A)$, pre-calibration). (a) A person standing still near a stairwell with a large backpack for an extended dwell, inside a labeled anomaly window: LOITERING fires at full strength and the fused belief correctly clears 0.5. (b) A person standing at an information board outside any labeled anomaly window – an everyday scene Avenue does not call anomalous: LOITERING still fires (dwell time alone cannot distinguish the two scenes), but a partially degraded skeleton ($r_{\text{pose}} = 0.834$, from viewing angle and partial occlusion) discounts the belief from 0.828 (what the undiscounted, –M1 pipeline would output) to 0.674 – the mechanism in §IV-B–§IV-C acting on a real frame, not a synthetic one. Both frames carry non-trivial conflict $K \approx 0.10$, reflecting that LOITERING is the only cue active in either case.

on an ordinary scene Avenue does not label anomalous: the same LOITERING cue fires at full strength as in panel (a), but a partially degraded skeleton discounts the belief from 0.828 (what an undiscounted pipeline outputs on this exact frame) to 0.674. We note this frame does not fully cross to a suppressed alarm at the default threshold – an honest limit of what one still-active-cue example can show; the aggregate effect of M1 across a whole stream is the controlled-noise result in §IV-E results ($19.2 \rightarrow 0.0$ FAPH), which is the claim this section illustrates rather than replaces. In both frames LOITERING is the sole active cue, so conflict K stays modest (≈ 0.10); we did not find a frame with two disagreeing cues active simultaneously on the Avenue test set, which is itself consistent with Table VI’s reading that Avenue’s real anomalies mostly fall outside our five cues’ coverage.

VII. LIMITATIONS

Reliability weighting is partly preceded. The reliability-aware prototype method [22] weights one model’s score by keypoint confidence; our novelty is discounting across a fused heterogeneous cue set with a conflict term and the downstream calibration and guarantee, not the idea of using confidence. *Normal-stream hours.* M4 needs enough confirmed-normal video; several pose-VAD benchmarks are short on it, so we report FAPH per available normal-hour with intervals and mark budgets that cannot be certified. *Cue coverage.* The five cues target human-action anomalies; events such as cycling or vehicle intrusion fall outside them, as for [24]. The fusion layer is cue-agnostic and could take on additional cue types without redesigning M2–M4, but we have not built or evaluated any. *Fixed cue thresholds do not appear to be the bottleneck.* The five cues’ thresholds (e.g. `loitering.dwell_seconds`,

crowd.threshold, fight.proximity) are fixed constants, tuned once rather than adapted per site; we built and tested a scene-adaptive variant that learns them from a site’s own confirmed-normal footage statistics only (calm/scene_autotune.py, no anomaly labels needed, so it is deployable the same way M4 self-calibrates). Three of five benchmark releases have no fully-normal clip to learn from at all; on the two that do, the learned thresholds moved substantially but changed frame AUC by +0.0005 (UBnormal) and −0.0053 (NWPU-Campus), with event F_1 unchanged in both. We report this honestly rather than claim a win: whatever limits detection on these benchmarks, it is not simply that our threshold constants are mistuned. *Exchangeability*. Learn-then-Test assumes the calibration and deployment normal streams are exchangeable; domain shift can break the guarantee, which we treat as a measured outcome rather than an assumption. *Hardware*. Every latency number in this paper is measured on one commodity laptop CPU (Intel Core i5-6300U); we have not measured on a genuinely embedded-class device (e.g. a Jetson or Raspberry Pi), so the decision layer’s edge-deployability is currently evidenced on desktop-class CPU only. *No operator study*. Part of this paper’s motivation is that uncalibrated alarms erode operator trust (§I); we have not run a study with real operators to confirm that a calibrated probability and an evidence record actually change trust or response behaviour in practice – the outputs are designed for that purpose, but the effect is not yet measured, on this system any more than on the prior work we compare against. *Real-data evidence for M1 is limited*. M1’s one demonstrated benefit (§VI-B) is a controlled result on a synthetic corrupted-skeleton stream. On the four real datasets we checked for a matching real-data effect, naturally low-reliability frames rarely coincide with an active cue (under 25% of active-cue frames, on every dataset and cutoff we tried), and when they do it is almost always inside footage already labelled anomalous rather than the confirmed-normal streams M1 is meant to protect; the largest well-powered real subset we could isolate (CHAD, low visible-joint-fraction frames with an active cue, $n=1355$) shows no benefit ($\Delta\text{AUC} = -0.041$). We report this rather than let the synthetic result stand in for a real-data one it has not yet earned. *No threshold reaches 0.9 event recall on any of the five benchmarks* (Table VI); the highest-recall FAPH cells are reported as – rather than filled in, so the false-alarms-per-hour claim in this paper is demonstrated at the recall levels our sweep actually reaches, not at the higher-recall operating point a deployment would likely prefer. We checked whether this is an artifact of the threshold grid rather than the data: sweeping down to the most permissive threshold possible (every frame alarmed, not just our usual grid’s floor of $\tau=0.02$) still only recovers 16.7% (Avenue), 45.1% (NWPU-Campus), 48.3% (ShanghaiTech), 68.8% (CHAD) and 80.2% (UBnormal) of ground-truth events – a real ceiling set by how many labelled anomalies our five cues ever engage with at all, not a search-range limitation (calm/recall_ceiling_check.py). *CALM-VAD’s own detection accuracy does not exceed B0 by a meaningful margin on any of the five benchmarks* (frame AUC

delta −0.005 to +0.003, Table VI and per-dataset reports): the calibration and risk-control gains in this paper are gains in what the fused belief means and how its alarm rate is bounded, not in ranking quality, and should not be read as evidence CALM-VAD out-ranks a plain weighted sum. *Not a leaderboard result*. CALM-VAD targets reliability and deployability; it is not proposed as a new frame-AUC state of the art.

VIII. CONCLUSION

Deployable edge video anomaly detection fails at its back end, not its front end: a black-box score and a hand-tuned threshold give operators neither a usable probability, an auditable rationale, nor a bound on false alarms. CALM-VAD replaces that back end with four lightweight stages—front-end reliability estimation, reliability-discounted evidence fusion, calibration, and distribution-free alarm-rate control—and emits every alert as an evidence record. Evaluated under a deployment-faithful protocol rather than frame AUC on five real benchmarks, it produces a calibrated incident probability with 56–95% lower calibration error than the raw fused belief on every one of them. A controlled study isolates the reliability mechanism, removing false alarms caused by front-end degradation entirely ($19.2 \rightarrow 0.0/\text{h}$) without a detection cost. And given roughly an hour of confirmed-normal calibration footage – supplied deliberately on UBnormal or found already sufficient in NWPU-Campus’s own split – it certifies an operator-chosen alarm budget with a distribution-free guarantee on two independent benchmarks, the first time this has been demonstrated for a heuristic multi-cue video anomaly detector, all at a CPU cost of about 0.1–1.5 ms per frame, independent of the dataset. Detection quality itself is honestly uneven, tracking how much of each benchmark’s anomaly taxonomy the five underlying cues were built to see. We take the resulting chance-level results on Avenue and NWPU-Campus – reached for two different, individually verified reasons – as evidence the method is not overclaiming, not as a failure to hide. The same two benchmarks are also where cross-dataset calibration transfer is worst (Table IX), a consistent picture rather than an isolated result. Code and the evaluation harness are released to make the protocol reusable, and to make the evaluation itself complete: all 20 cross-dataset pairs across all five benchmarks are included.

REFERENCES

- [1] H. Le, C.-K. Lu, and C.-C. Hsu, “Video anomaly detection for edge-based IoT systems: A survey of input modalities and real-time applications,” *Intelligent Systems with Applications (Elsevier)*, vol. 29, 2026.
- [2] Y. Liu, S. Liu, X. Zhu, J. Li, H. Yang, L. Teng, J. Guo, Y. Wang, D. Yang, and J. Liu, “Privacy-preserving video anomaly detection: A survey,” *arXiv preprint arXiv:2411.14565*, 2024.
- [3] MarketsandMarkets, “AI video analytics for smart cities market – global forecast to 2030,” Market research report, 2025.
- [4] M. Rashidi, “Benchmark AUC is not deployable reliability: A cross-dataset audit of off-the-shelf features for surveillance video anomaly detection,” *arXiv preprint arXiv:2606.29506*, 2026.
- [5] N. Rashvand, S. Yao, A. Danesh Pazho, B. Rahimi Ardabili, and H. Tabkhi, “From frames to events: Rethinking evaluation in human-centric video anomaly detection,” *arXiv preprint arXiv:2604.09327*, 2026.

- [6] S. Tariq, M. B. Chhetri, S. Nepal, and C. Paris, "Alert fatigue in security operations centres: Research challenges and opportunities," *ACM Computing Surveys*, vol. 57, 2025.
- [7] N. Dadashi, A. W. Stedmon, and T. P. Pridmore, "Semi-automated CCTV surveillance: The effects of system confidence, system accuracy and task complexity on operator vigilance, reliance and workload," *Applied Ergonomics*, vol. 44, no. 5, pp. 730–738, 2013.
- [8] F. Donald, C. Donald, and A. Thatcher, "Work exposure and vigilance decrements in closed circuit television surveillance," *Applied Ergonomics*, vol. 47, pp. 220–228, 2015.
- [9] G. Alinezhad Noghre, A. Danesh Pazho, V. Katariya, and H. Tabkhi, "Understanding the challenges and opportunities of pose-based anomaly detection," in *Proc. 8th Int. Workshop on Sensor-Based Activity Recognition and Artificial Intelligence (iWOAR)*, ACM, 2023.
- [10] I. Song and J. Lee, "Bounding-box trajectories matter for video anomaly detection," *arXiv preprint arXiv:2605.21957*, 2026, to appear, ECCV 2026.
- [11] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *arXiv preprint arXiv:2110.01052*, 2021.
- [12] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. I. Jordan, "Distribution-free, risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, 2021.
- [13] G. Segal, N. Keidar, R. M. Lotan, Y. Romano, M. Herskovitz, and Y. Yaniv, "Utilizing risk-controlling prediction calibration to reduce false alarm rates in epileptic seizure prediction," *Frontiers in Neuroscience*, vol. 17, p. 1184990, 2023.
- [14] S. Yuan, J. Li, C. Wang, and X. Zhang, "Conformal machine learning for reliable anomaly detection in industrial cyber-physical systems," *Reliability Engineering & System Safety (Elsevier)*, vol. 274, 2026.
- [15] European Parliament and Council, "Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act)," <https://artificialintelligenceact.eu/>, 2024.
- [16] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," <https://github.com/ultralytics/ultralytics>, 2023.
- [17] Y. Zhang *et al.*, "Bytetrack: Multi-object tracking by associating every detection box," in *Proc. European Conference on Computer Vision (ECCV)*, 2022.
- [18] O. Hirschorn and S. Avidan, "Normalizing flows for human pose anomaly detection," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [19] A. Karami, T. K. K. Ho, and N. Armanfard, "Graph-jigsaw conditioned diffusion model for skeleton-based video anomaly detection," *arXiv preprint arXiv:2403.12172*, 2024.
- [20] A. Delic, M. Grcic, and S. Segvic, "Sequential keypoint density estimator: an overlooked baseline of skeleton-based video anomaly detection," *arXiv preprint arXiv:2506.18368*, 2025.
- [21] C. Tang, S. Zhou, H. Shi, and L. Wang, "Action hints: Semantic typicality and context uniqueness for generalizable skeleton-based video anomaly detection," *arXiv preprint arXiv:2509.11058*, 2025.
- [22] N. Dong, Y. Su, X. Dong, Z. Jiao, X. Guo, and Z. Pan, "Reliability-aware prototype calibration for frozen pose-flow video anomaly detection," *arXiv preprint arXiv:2606.20312*, 2026.
- [23] J. Liu, Y. Liu, J. Lin, J. Li, L. Cao, P. Sun, B. Hu, L. Song, A. Boukerche, and V. C. M. Leung, "Networking systems for video anomaly detection: A tutorial and survey," *ACM Computing Surveys*, 2024, *arXiv preprint arXiv:2405.10347*.
- [24] V. G. Warnasooriya, A. Hajian, W. Ruangsang, and S. Aramvith, "Real-time video anomaly detection using YOLO pose estimation and CLIP-based semantic scoring," *arXiv preprint arXiv:2608.31074*, 2026, also: *Proc. 9th IEEE Int. Conf. on Multimedia Information Processing and Retrieval (MIPR 2026)*.
- [25] M. Ye, W. Liu, and P. He, "VERA: Explainable video anomaly detection via verbalized learning of vision-language models," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [26] K. S. Yadav and L. Kumar, "Transformer-driven multimodal fusion for explainable suspiciousness estimation in visual surveillance," *arXiv preprint arXiv:2512.09311*, 2025.
- [27] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," *Advances in Large Margin Classifiers*, 1999.
- [28] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," *Proc. ACM SIGKDD*, 2002.
- [29] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. International Conference on Machine Learning (ICML)*, 2017.
- [30] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using bayesian binning," in *Proc. AAAI Conference on Artificial Intelligence*, 2015.
- [31] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.
- [32] G. Shafer, *A Mathematical Theory of Evidence*. Princeton University Press, 1976.
- [33] P. Smets and R. Kennes, "The transferable belief model," *Artificial Intelligence*, vol. 66, no. 2, 1994.
- [34] D. Yu and D. Frincke, "Alert confidence fusion in intrusion detection systems with extended dempster-shafer theory," in *Proc. 43rd ACM Southeast Conference*, 2005.
- [35] W. Luo, W. Liu, and S. Gao, "A revisit of sparse coding based anomaly detection in stacked RNN framework," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [36] A. Acsintoae *et al.*, "UBnormal: New benchmark for supervised open-set video anomaly detection," in *Proc. IEEE/CVF CVPR*, 2022.
- [37] A. Danesh Pazho *et al.*, "CHAD: Charlotte anomaly dataset," *Proc. Scandinavian Conf. on Image Analysis (SCIA)*, 2023.
- [38] C. Lu, J. Shi, and J. Jia, "Abnormal event detection at 150 FPS in MATLAB," in *Proc. IEEE ICCV*, 2013.
- [39] C. Cao, Y. Lu, P. Wang, and Y. Zhang, "A new comprehensive benchmark for semi-supervised video anomaly detection and anticipation (NWPU campus)," *Proc. IEEE/CVF CVPR*, 2023.